June 12, 2025

Dr. Mehmet Oz
Administrator
Centers for Medicare & Medicaid Services
Department of Health and Human Services
Attention: CMS-0042-NC / RIN 0938-AV68
Submitted electronically to: http://www.regulations.gov

Dear Administrator Oz:

We applaud CMS for taking a leadership role in ensuring that patients, along with many other entities in the health system, are able to leverage digital technologies to access the information they need to make good healthcare choices.

Below we provide more detailed answers to questions posed in the RFI.

We welcome the opportunity to engage further with CMS on this subject – please do not hesitate to contact Eric Ellsworth at eellsworth@checkbook.org or 202-454-3010 if you have any questions or we can provide further information or assistance.

Sincerely,

Eric Ellsworth

Director, Health Data Strategy

Consumers' Checkbook/Center for the Study of Services

CONSUMERS' CHECKBOOK

CSS Center for the Study of Services

Contents:

Organizational Background 2

Accountability and Oversight Needed for Accurate Provider Directories 3

Accountability mechanisms for consumer-centered digital workflows. 6

Need for a Uniform Consumer-Facing Network Identifier 7

Need for Standards in Formulary Data and Search Mechanisms 9

- Identity of the formulary 9
- Searchable scope of the formulary 10
- Explicit assertion of non-coverage 11
- Codeset conversion 11
- Representing policy information from formularies 12
- Recommendations for Improvement of Formulary Data and Search Experiences 13

Need for Standardization in Insurers' Prior Authorization and Coverage Rules 14

- Problems with Documentation and Use of Medical Necessity and Coverage Criteria 14
- How These Problems Affect Prior Authorization 16
- Potential Improvements for Prior Authorization and Insurer Coverage Rules 16

Organizational Background

Center for the Study of Services (d/b/a Consumers' Checkbook) is a nonprofit organization whose mission is to help consumers make educated choices of service providers. For over 40 years, we have provided information to help consumers select health insurance plans, medical providers, and navigate many other complexities of the US health system. We are committed to ensuring that consumers have the information they need to make good choices about their healthcare.

Checkbook's plan comparison tools have been used by hundreds of thousands ACA consumers in numerous state exchanges, by millions of federal employees, and by tens of thousands of consumers shopping on private exchanges. This has given us intimate familiarity with the benefit design of ACA plans, as well as on-the-ground experience with how data on plan design, and related data such as provider-network participation, is collected and managed. We have also built Medicare Advantage plan choice tools for Medicare beneficiaries with End State Renal Disease, including estimating of out-of-pocket costs for both medical services and prescription drugs focused on typical ESRD utilization patterns. We have worked with Medicare Advantage plan design data and several different sources of that data (Plan Finder PUF, Medicare Landscape File, PBP Benefits files) and are familiar with benefit designs of Part C and D plans.

Checkbook has decades of experience helping consumers shop for insurance plans and health providers, and this has given us insight into how coverage questions affect patients and consumers. In the course of this work, we have observed many problems associated with payment rules and coverage of services.

We have long advocated for transparency and accessibility of information consumers need to make educated choices of healthcare services and service providers.

The Center's work also includes evaluations of quality of insurance plans, medical groups, individual providers, dialysis facilities, , and many other entities through surveys of plan members, patients, and healthcare providers. We also use claims data and other administrative data from Medicare, Medicaid, and Marketplace sources to assess the quality of various entities..

Accountability and Oversight Needed for Accurate Provider Directories

RFI questions addressed: TD-5 TD-1,PR-7, PC-1

Our organization has decades of experience helping consumers choose high quality providers and understand coverage of their providers when choosing an insurance plan. In the course of this work, we have observed many of the problems outlined in the 2022 RFI CMS-0058-NC and have observed the adverse consequences for patients of inaccurate provider directories. Inaccurate provider directory data constitutes one of the greatest frustrations for consumers in both shopping for and accessing care, and hampers the development of effective shopping experiences.

Of the many problems with provider directories and limitations in information consumers have on providers, we believe that the most pressing one is making available accurate and up-to-date information on individual provider-organization affiliation that can be used to determine accurate in-network status for any individual provider. The absence of timely and accurate provider-organization affiliation data is the primary driver of the inaccuracy observed in provider directories, as well as the primary obstacle to understanding the capacities of those organizations. The lack of organizational affiliation data makes it difficult for insurers to purge outdated addresses for providers, leading to incorrect address records in data available to consumers. The lack of a reliable central record system for capturing this data is a primary factor driving the current fragmented system and the associated burden on providers of updating numerous insurers' provider portals regularly. As such, we suggest that CMS should prioritize efforts to capture and maintain accurate and up-to-date affiliation data.

From a policy perspective, we believe that the most important consideration in developing a multi-party system is ensuring that the rules governing the system hold each participant accountable for the data under their control. In particular, providers should be held directly accountable – independently of insurers - for making accurate and timely updates to a centralized system for data elements about affiliation of individual providers with organizations and any other business entity that contracts with insurers. We note that 45 CFR 162.410(a)(5) already provides authority for HHS to require providers to make timely updates to NPPES in the event of changes to data about them. Holding providers accountable in turn will permit holding

insurers accountable for the accuracy and timeliness of data that they control, such as contracts with provider groups and status of onboarding individual providers.

We further suggest that providers should be required to maintain accurate organization affiliation data in real-time. Since the source data for providers' organizational affiliations is contained within one or more information systems, it is eminently feasible for the developers of these system to create interfaces that allow for automated transmission of the appropriate data . Automated jobs of this nature, combined with careful data validation requirements would both reduce the administrative burden on provider organizations and result in much improved accuracy of the data collected. While such a requirement might require more work up front, the benefits both for consumers and in terms of designing out complex versioning problems related to stale data would be overwhelmingly valuable.

There are many steps involved in the collection and use of provider data throughout the health system. Because both providers and insurers are likely to rely on intermediaries for these steps, we suggest that CMS' efforts be focused on ensuring that all providers can meaningfully make use of an intermediary to submit to the NHD and can readily consume data in an NHD. That includes the development of standards for key metadata (the timing of data collection from providers, sources from which the data is collected, etc.) as well as rules ensuring that such intermediaries cannot block the public release of data collected in the NHD. The ability to use intermediaries, coupled with high degree of visibility into the process of gathering and cleansing the data, will be pivotal for ensuring that the NHD succeeds in reducing burden on providers and insurers and generating a consolidated set of high quality data on providers and organizations (meaning legal entities through which providers do business).

In order for an NHD to avoid becoming another data silo encompassing a subset of providers, it must be able to meet the needs of multiple stakeholders throughout the health system. To meet this need it must be built on a data model that handles many distinct types affiliations between individual providers and organizational entities as well as between various organizational entities, and establish clear and permanent identifiers for organizations, sub-organizations and other business entities. It is critical that these organizational entities be properly defined and their identifiers with the NHD align with the identifiers used in insurer contracts with entities involved in care delivery. In particular, either TINs or EINs used in contracts should be disclosed publicly, or where this is not feasible, a consistent, immutable pseudo-TIN identifier should be generated and used. We also note that the lack of a consistent identifier for hospitals hampers many use cases for many stakeholders in the industry.

An NHD may require CMS to understand and map both business relationships between various types of provider group entities, such as distinct radiology practices that were acquired by an investment group and owned by a holding company. Business relationships of this type may be incorporated into different insurer contracts in different forms or over time. For example:one

insurer at one time may contract with each radiology group individually, while another insurer at another time might contract with the investment group for some of the practices, while still another might contract only with the holding company. Such mappings will need to be maintained over time, and therefore also require clear accountability mechanisms to ensure that the data is accurate and up-to-date. The incentives for disclosing such data may vary according the parties involved, and some parties may resist such disclosure and delay the collection process. As such, we suggest that HHS invest energy early on in assessing which authorities and regulatory mechanisms will be needed to ensure that accurate data is collected and maintained.

Efforts to develop a sufficient identification scheme should be undertaken in an open forum with participation of relevant stakeholders, and be structured to create an identifier system that fully represents the distinct kinds of organizational entities that exist and the relationships between them. Further oversight will be needed to ensure that the entities named within insurer – provider contracts align with the organizational identifiers that are defined by HHS or others. Another critical element for success will be ensuring that full understanding and consideration of the workflows of insurers, provider groups, and other stakeholders who convert these provider-insurer contracts into used in provider directories.

We further note that many licensing, credentialing and other processes both refer to individual providers and organizational entities, and produce data that should be incorporated into the NHD workflow. These processes vary greatly between states, and include state government entities, local government entities, and other private entities that may be involved in licensing. Our experience as users of state licensure data is that these entities appear to use legacy IT systems that may be difficult to upgrade. Many licensure datasets do not use NPIs as identifiers of providers, and suffer from poor data quality. Criteria and timelines for renewal, data definitions, and types of data all vary between licensing entities. We suggest that HHS expedite collection of licensure data by creating standards for licensure and potentially offering funding to licensure boards for upgrades to their IT infrastructure.

A high-quality directory of providers is an essential component of successful and efficient functioning of the world's costliest and most complex health system. Based on our experience the current data structures, collection processes, and oversight NPPES are insufficient to meet this need. The data in NPPES – particularly data related to organizations and relationships to individual providers – is not sufficiently well-defined or validated to meet the needs of consumers and many other health system stakeholders. To maximize its likelihood of success, , both development and maintenance of an NHD should be done under transparent and open governance structures. Given CMS' commitment to FHIR and regulatory references to these standards, FHIR workgroups would be a useful resource for informing the development of an NHD. However we caution that the proliferation of FHIR workgroups, and the pace at which they have been mandated to operate, has resulted in a large body of code and documentation

that is relatively inaccessible without a steep learning curve. Given the central role of the NHD in the functioning of the US health system, we suggest that there must be greater efforts to ensure that assets are readily comprehensible and usable for all parties, including the public.

Accountability mechanisms for consumer-centered digital workflows.

RFI Questions Addressed: TD-4,TD-1,PC-1, PC-5, PC-7, PC-12

When consumers interact with the health system, their end-to-end workflow involves interactions with multiple entities within the health system. The complexity of these interactions and the poor integration between them is a central part of consumer frustration with the healthcare system:

"The No. 1 complaint from customers is the healthcare experience is not integrated," Tilak Mandadi, CVS Health's chief experience and technology officer, told the publication."

Despite the critical role that reducing consumers' friction and administrative burden plays in delivering value and helping consumers achieve better outcomes, there is little detailed measurement of friction or administrative burden in consumers' healthcare workflows.

Experienced technologists affirm that the quality of a digital experience depends as much or more on the design and testing that went into creating it as it does the on the technology stack used. Even a cursory look at many of the consumer-facing workflows shows a wide variation in the quality of the user experience. Some digital workflows embody the promise of interoperability – notably some excellent EHR deployments that produce an integrated EOB/bill consumers can actually understand and act upon. Yet many, if not most, consumer workflows, such as providers' patient intake forms, consumer-facing information on coverage or prior authorization rules, or determining if a pharmacy has a needed supply in stock, are poorly designed and have gaps that leave the consumer unable to understand what to do or solve their problem. In the worst cases, when poorly designed digital workflows have completely supplanted human-driven workflows but are poorly tested, the consumer not only cannot traverse the digital workflow, but they also cannot reach a human to get help.

Software engineers know the importance of system integration testing to ensure that the pieces of complex systems successfully work together, and that changes to one piece don't break another piece. Integration testing is especially important when a workflow involves the use of different systems with different system owners; without it, workflows that cross multiple systems fail, causing chaos in the workflows and wasting huge amounts of time finding and fixing the source of the failure.

Consumer workflows very frequently cross between different systems controlled by different businesses, and as in software engineering system integration problems are legion.

Friction problems are self-reinforcing – once a consumer has found some setup that works, the cost and effort to do anything else is prohibitive. When consumers and others face high costs to switch service providers, this reduces industry competition and keeps total costs high. And unlike buying cars or TVs, consumers who choose the wrong provider or type of care cannot reverse these choices – there are no returns or resales.

Without incentives or accountability to develop seamless, low-friction experiences – i.e. "integration testing" - the interoperability tech stack alone cannot solve the problems of disconnected and poorly integrated consumer workflows. In order for HHS to meet its goal of empowering consumers, CMS will need to develop incentives and accountability structures that reward market actors for developing higher quality, less burdensome experiences.

CMS has recognized the importance of measuring and incentivizing quality across all its payment programs, through Star Ratings programs for Medicare Advantage Plans, Hospitals, Nursing Homes, as well as payment programs like Quality Bonus Payments, Medicare Shared Savings Programs, Merit-based Incentive Payments System, and others. This has given the agency experience with developing and evaluating measurement frameworks, as well as collecting and reviewing actual quality data, and tackling complex problems of attributing good or bad outcomes to particular people within the overall care workflows.

There are important measures of patient experience in use within these programs today, such as CAHPS surveys, these measures focus more on the patient's ultimate success or failure in obtaining services (e.g. accessing care when needed), the ease of navigating the system, and their overall experience with the provider or the insurer. While these measures may get at the complexity of navigating the system, the questions are difficult to attribute solely to provider organizations and make it challenging to assess an entity's performance in minimizing consumer friction.

Interoperability technologies, combined with Human Centered Design techniques, can be leveraged to measure both the degree of difficulty in consumer workflows, as well as identifying specific failures that are costing consumers effort and stymying their access to care. Companies in many other industries leverage these techniques and integration of data to simplify the customer journey. Consumer tech platforms such as Uber or Apple carefully map out consumer journeys and definitions of success across these journeys. They monitor "conversion metrics" from one step of the journey to another to identify friction and points where consumers get stuck. Applying these techniques to consumer workflows can similarly illuminate friction and points of failure.

We suggest that HHS develop and implement consumer-centered measures that capture the administrative friction consumers experience in their care journey. Such measures can form the basis of incentive payments to ensure that the parties deploying and using the interoperability tech stack undertake the necessary design and testing to create consumer workflows that truly achieve HHS' vision of empowering consumers to control their own healthcare.

Need for a Uniform Consumer-Facing Network Identifier

RFI questions addressed: TD-4, TD-5,PC-4, PC-5, PC-6, PC-8,

Provider networks play a pivotal role in the US healthcare system. Since most insured Americans cannot afford to access medical care without using their insurance, they incentivized to receive care from providers and facilities that are "in-network" under a set of rules – including which benefits are covered, policies about which services insurers will pay for and

under what circumstances (i.e. medical necessity policies), payment rates, etc.- that are established by the contracts that define each provider network. In short, much of what determines a patient's experience in the provision and payment of healthcare is set at the level of the provider network.

In spite of their central role these networks are never clearly identified, either in public consumer-facing materials such as insurers' websites, plan brochures, or in datasets such as provider-network data files, price transparency data, policy reports, consumer experience surveys like CAHPS, or almost any other data reported publicly or privately. There is no consistent name or number that a consumer can find to understand what network they are in or allow them to see if the same network is available through a different health plan or insurer (which is usually the case).

The lack of a well-defined network makes numerous aspects of consumer choice extremely difficult. When shopping for health plans, Medicare and Marketplace consumers are presented with a myriad of plan choices but cannot meaningfully discern how different plan options would affect their access to care through the network provider. When trying to find an in-network provider, they face names and identifiers that do not match their plan card or other materials.

Despite the large number of health plans offered to consumers in markets such as ACA exchanges or Medicare Advantage plans, most insurers operate a relatively small set of networks upon which these plans are built. The current lack of consistent identification forces consumers to traverse the complex, messy, and changing details of plans as a way to learn about costs, in-network status of providers, and coverage rules that in reality are defined at the level of the network.

We propose that CMS develop a standardized system for identifying provider networks, and require carriers to use these identifiers in public-facing assets such as provider directories, plan cards, websites, plan brochures, as well as in data sources like FHIR provider directory APIs and Transparency in Coverage price files. With this standardized identifier, application developers could link together the necessary data to permit consumers to understand and make educated decisions about the costs and coverage implications of choosing different providers and services.

Attached to this response are number of materials demonstrating the need for such identifiers, benefits they would bring, and potential statutory authorities for such an identifier.

CONSUMERS' CHECKBOOK

CSS Center for the Study of Services

Need for Standards in Formulary Data and Search Mechanisms

RFI questions addressed: TD-4, PC-4, PC-5

Checkbook's deployment of multiple formulary search tools, our extensive work with machine-readable formulary data, review of numerous paper/PDF formularies, and participation in the design of the FHIR Formulary resource within the FHIR PDEX workstream has led us to observe numerous problems that hamper consumers' ability to get actionable information about coverage for medications they need.

Currently formulary data suffers from data quality problems which can result in inaccurate information for consumers, as well as frustration and additional costs for others within the healthcare system in dealing with erroneous information. Data quality issues are pervasive in consumer-facing administrative datasets, and while the poor accuracy of provider directories data has been widely discussed by both the policy community and the public at large, much less scrutiny has been directed at the accuracy of formularies.

We believe CMS and ONC/ASTP's rulemaking and enforcement should address data quality problems in formulary lookups. Some key problems we believe merit attention:

1- Ensuring the identity of the formulary – i.e., that the formulary put forward is clearly associated with appropriate health plan options relevant to individual consumers.
2- Clearly defining the intended "searchable scope" of the formulary – i.e. what drugs are covered and the coverage terms.
3- Requiring the formulary to affirmatively assert non-coverage of a drug instead of only asserting coverage and leaving non-coverage to be inferred via absence from a list of covered drugs.
4- Ensuring that codeset conversion (e.g. NDC to RxCUI) does not result in inaccuracies.
5- Ensuring that FHIR resources correctly capture all of the policy information contained in formulary documents.

We discuss each of these further below; additional description and examples can be found in the attached document "Proposed changes to resource definitions for DaVinci PDEX Drug Formulary".

Identity of the formulary

Currently formularies do not a have a public identifier. In public insurance markets such as Medicare Advantage or the ACA Marketplaces, insurers use the same formulary for numerous plans. In fact, in our review we have found that nearly all ACA plans in a given state use one or occasionally two formularies. However, these formularies are not assigned their own public identifier analogous to Medicare Advantage plan's "H number" or a Marketplace plan's "HIOS ID". As a result, when consumers search for formularies via web searches, insurers' websites, or other open data resources, it is often unclear whether they are looking at the right formulary. In some cases, insurers create links to plan-specific online formulary lookup tools, but these are also generally not labeled with a plan or formulary identifier, making it very difficult to independently check for errors in the data or links that are incorrectly affiliated with a HIOS ID.

If a consumer is an existing member of an insurance plan, and they login to the insurers' website or app, in theory that consumer will be automatically directed to the correct formulary. However, this approach limits the ability of third parties to create apps and websites for consumers, since getting the consumer to the correct formulary requires a login. Shopping for health plans is particularly limited by this approach, since the consumer is not yet a member of plans s/he may wish to consider.

It is worth noting that formularies within the ACA and Medicare Advantage markets do have **non-public** identifiers. Drug formulary and cost data on Medicare Part D and Part C plans have formulary identifiers which are published as part of public use files. However, when the data is published through another means (such as a PDF posted on an insurers website, a website such as Formulary Navigator, or ultimately a FHIR Formulary resource), there is no consistent use of this identifier. Nor is the identifier is published to consumers in plan brochures, on their insurance cards, etc. As such consumers cannot use this identifier to find their own formulary or confirm that the formulary they are looking at is in fact the relevant one for them.

In the ACA markets, SERFF data contains an identifier for a formulary, but these identifiers are not global - i.e. they take the form TXF001 within a particular SERFF binder, but TXF001 is used by each issuer and may even be duplicated within the same issuer if they file multiple SERFF binders. As such this identifier cannot be uniquely determined. Similar to Part D and Part C plans, no formulary identifier is published to consumers or used on third party websites.

Searchable scope of the formulary

When an application goes to search for a drug using a FHIR Formulary resource (or any other search interface), the client application will send either a code representing a drug or a string representing a drug name. The most common arrangement in our experience is for the application to use string search in "typeahead" format to provide a list of drug names, then ask the user to click on the drug of interest (which often includes dose form and strength). Clicking on the drug converts the name to a code such as RxCUI or NDC. Then client then sends this code to the API, which searches the code against a list of covered drugs. The API/FHIR resource will return a list of covered drugs that match either the code or a text search term.

If drug code is not found within list of drugs for the formulary, nearly all client applications interpret this to mean that the drug is not covered. However, this conclusion is questionable, as it is contingent upon the assumption the drug to be searched was in the scope of the formulary - i.e. that the formulary purported to have information about whether this drug was covered or not. If the drug was intended to be within the search scope, then it is correct to infer that the drug is not covered. If, however, the drug was not part of the search scope, then the correct inference is "not in scope"; inferring non-coverage is a false negative result.

Of the numerous drug search interfaces we have reviewed, none explicitly defines the search scope. Yet in review of "paper" formularies (i.e. formularies aiming to be read by a person), most of them have scope-defining text that describes coverage under this formulary (which functions as an annex to a plan contract). That text may note that specialty drugs are out of scope of the formulary and are covered under a different formulary, or it may note blanket coverage of some classes of medication (e.g. various types). Physician-administered or "medical service" drugs are also generally not addressed via these

formularies (and in fact most insurers do not make available any consumer-facing documentation on coverage of such drugs). The written text of a formulary may also note blanket non-coverage of medications such those for alopecia or erectile dysfunction medications.

Because of the lack of an explicit search scope, client applications must decide for themselves what constitutes an appropriate "search universe" from the large number of drugs in standard codesets like NDC or RxCUI. If an application developer decides on a restrictive scope, some drugs that are on a given formulary may not even be searchable, because they removed from the search universe. Converse if the application developer uses a large search universe it is likely to generate false negative coverage results because drugs in the search universe are outside of the search scope.

Explicit assertion of non-coverage

One simple step that would reduce avoid false negative search results is to require that machine-readable formulary data list both covered and non-covered drugs. If application developers had a list with both covered and non-covered drugs, their applications could correctly determine "not in scope" results and avoid false negatives. When comparing multiple formularies (as in plan shopping uses cases) this would likely still result in some frustrating results, as it may be necessary to show "Drug Not Listed" side by side with "Covered" or "Not Covered" results, which effectively stalls the consumers' comparison of plans or formularies.

In public insurance markets such as Medicare Advantage and ACA marketplace, if non-covered drugs were listed, regulators could review submitted machine-readable formulary lists for differences in the scope of the list and take action to harmonize these lists.

An additional step that could be taken is to extend the FHIR Formulary resource so that it provides linkage to other formularies that do or may list the out of scope drug. For example, if an insurer has distinct formularies for specialty drugs vs non-specialty drugs, the Formulary resource could direct the client to the appropriate Formulary resource for the drug which was queried.

Codeset conversion

Most formularies are created and maintained by insurers (or more often their pharmacy benefit manager partners) using NDC codes to identify each drug. This is because negotiated pricing of drugs is generally per-NDC, with variations in price by manufacturer and sometimes dose form. However, in presenting formularies to consumers it is often necessary to represent coverage of drugs at a less detailed level than individual NDCs. Sometimes formularies are assembled and presented by organizing drugs via classes and then using generic or common names to list coverage. In other cases, formularies use RxCUI codes to map NDCs to higher level drug concepts. Both ACA Marketplaces and Medicare Advantage formularies specify the use of RxCUIs to indicate coverage or non-coverage. As such producing formulary data for regulators (which is then downloaded and used by app developers, researchers, and others), insurers must cross-walk NDC codes to RxCUIs. There are a number of issues that arise from this crosswalking.

First, it is unclear how reliable the NDC-RxCUI mapping generated by RxNorm is. The NDC codeset is one input into RxNorm's semantic mapping algorithm, but it is unclear to us whether the NDC-RxCUI mappings have been tested for reliability or exhaustiveness. We are not aware of any other NDC-RxCUI mapping than the one created by RxNorm.

Second, the terminology and meaning of "archived" drugs within RxNorm may not match formulary development or use in coverage. It appears to us (though we are not aware of definitive testing in this case), that the removal of an NDC code from FDA's daily published list of NDC codes is a primary trigger for changing the status of RxCUIs associated with this NDC from "active" to "archived". This creates a dilemma for application developers in determining the appropriate search universe of RxCUIs. Most archived RxCUIs are no longer in clinical use and should properly be excluded from the search universe presented to users. However, in some cases NDCs that have been removed from the FDA's list are still be present in the supply chain and available at the pharmacy.

There is also inconsistency in formulary data with regard to inclusion of new drugs. In Marketplace plans, this occurs because the data collection occurs over a year before the plans start coverage. This results in both inconsistent listings of new drug RxCUIs and version issues with regards to which release of RxNorm should be used.

Third, in some cases NDC codes that map to the same RxCUI but are listed on different tiers of the formulary. This generally occurs because the NDC-RxCUI mapping grouped together NDCs that may have different costs. For example, if a drug is available in both a sublingual dissolvable strip and another dissolvable form, both of these could map to the same RxCUI. If the sublingual dissolvable strip is priced much differently from other dissolvable forms, then it is likely to be listed on a distinct formulary tier. However, most formulary data collection mechanisms involving RxCUI lack any way to distinguish these two cases. As such if that RxCUI is searched, it will inexplicably be listed on two distinct tiers, for reasons that the user cannot know. Additionally, the presence of duplicate entries causes significant problems with data quality check steps in populating formulary lookup databases. Most data loading steps involve joining multiple recordsets, and ensuring that these joins work correctly usually means avoiding duplicated RxCUIs. Without a clear way of delineating why a given RxCUI is listed twice, it is impossible to distinguish that occurrence from an error in either populating the machine-readable list or in downstream handling of the data.

Representing policy information from formularies

Within discussion of machine-readable formulary data, there is often an unstated narrowing of the definition of a formulary that can adversely affect interoperability. A formulary - in the US, other countries use this term differently – is an insurer policy document which discloses its coverage of different prescription drugs. This policy document is incorporated by reference into member and provider contracts with insurers respectively. To set the coverage of drugs, a formulary document performs a number of functions. One central function, listing specific drugs that are covered and establishing coverage tiers, has been the focus of interoperability efforts thus far. However, as referenced elsewhere in this comment, formularies also contain many other important kinds of policies

that set the terms for insurer payment of drugs under the health plans which reference that formulary. Those might include:

- Blanket statements of non-coverage **any** medications used to treat conditions such as alopecia (hair loss), erectile dysfunction, etc.
- Policies regarding payment of over-the-counter medications for particular health needs (e.g. prenatal vitamins, allergy medications)
- Policies on payment for medications that can be either pharmacy dispensed or over-the-counter (omeprazole, famotidine, etc.)
- Definitions, requirements, and related policy documentation around prior authorization, step therapy, etc.
- Statements about experimental use of medications
- Policy rules regarding compounding
- Requirements for use of specific pharmacies for particular types of medications
- References to other formularies (e.g. specialty drug formularies, etc.)

These types of information are not yet well-represented in machine-readable form. As described above, such policy terms and information may be very meaningful for consumers and/or providers when trying to understand coverage of medications within their current plan, a plan they are shopping for, or plans of people they are caregivers for.

Recommendations for Improvement of Formulary Data and Search Experiences

We believe recommend that ONC/ASTP and/or CMS:

- Require formularies to define or reference explicit declaration of search scope.
- Work to harmonize the search scope between formularies.
- Require the use of a standardized public identifier for each distinct formulary an insurer maintains.
- Require insurers to provide the formulary identifier to consumers on their member ID, within the Summary of Benefits and Coverage (SBC) document, contract info, plan brochures, on websites, etc.
- Require insurers to explicitly list non-covered drugs within machine-readable formulary data they use and disseminate.
- Perform more extensive testing of the NDC-RxCUI mappings generated by RxNorm to ensure that the mappings are adequate for use in creating and maintaining machine-readable formularies.
- Create mechanisms with machine-readable formularies to distinguish listings the same RxCUI on different formulary tiers. This might include capturing information on different NDCs that map to the same RxCUI, or different coverage reasons (i.e. distinct clinical indications) that might result in differences in tiering for the same RxCUI.
- Require FHIR Formulary resources and other machine-readable formulary implementations to capture additional information about a formulary sufficient to

completely define and compute the rules of coverage of a medication within the plans referenced by that formulary.

Need for Standardization in Insurers' Prior Authorization and Coverage Rules

RFI Topics Addressed:

Problems with Documentation and Use of Medical Necessity and Coverage Criteria

All health insurance claims are assessed, in some fashion, to determine whether the services rendered are considered medically necessary to treat a patient's medical concerns, and whether the insurer covers the products or services the provider has employed or will employ for the patient. As such, medical necessity and coverage criteria (which we'll call "payment criteria" in these comments) exist for every single service for which an insurer pays. Prior authorization processes add a layer on top of these criteria by requesting additional clinical information, with the nominal goal of assessing whether a service considered medically necessary in some circumstances is appropriate in a patient's own specific circumstances and providing authorization for the use of the service in advance of the patient receiving.

In general, each insurer develops their own set of payment criteria for all the services they cover, meaning that there are as many different sets of payment criteria as there are insurers. In fact, many insurers maintain distinct sets of payment criteria for different markets (e.g., Medicare vs commercial insurance vs Medicaid managed care). ERISA and the ACA and subsequent regulations require insurers to maintain written payment criteria, and indeed insurers do maintain various forms of documentation of payment criteria. However, the methods insurers use for documenting these criteria are burdensome for providers and essentially unusable for patients.

Currently, most insurers create some form of "provider manual" describing payment criteria as well as insurer-specific instructions for how to code medical records using codesets suitable for claims processing (e.g. diagnostic codes, procedure and other service codes, drug codes, admissions codes, and various forms of modifiers to each codes). In some cases, provider manuals are published in a PDF document that runs hundreds of pages; in other cases the insurer creates multiple distinct documents and posts them to a website. There is no consistency between insurers in any aspect of this documentation: neither the form of publication, nor which types are services are published together, nor how they are organized and indexed, are standardized at all. Many insurers adopt by reference other guidelines such as Milliman Care Guidelines, or Medicare Fee For Service local and national coverage determinations. They also reference quality measures and standards from organizations such as NCQA, guidelines from medical specialty societies, and the academic literature. Since these payment criteria documents are updated on timeframes ranging from months to years, quite frequently the external references to which these documents refer are revised after the document is written, and payment criteria documents often do not clarify which versions of the referenced materials are considered applicable.

These general payment criteria are used to assess medical necessity and coverage of every single claim that is paid by using automated rules to accept or reject the claim (sometimes called "edit checks") based on checking that the codes have been applied. As a result of these complex rules, a large

proportion of submitted claims (8-40%) are initially rejected as not meeting payment criteria and coding requirements. That initial rejection rate is staggeringly inefficient – imagine any 8-40% of credit card payment requests being declined because of transactional incompatibilities between the seller and the merchant bank for the buyer's credit card. Reducing this burden and inefficiency would dramatically improve the functioning of the health system for all parties.

Many physicians transact with 30 or sometimes even 50 distinct insurers as part of seeing their panel of patients. That means they must (in theory) maintain awareness of each of 30-50 insurers' distinct set of rules when considering whether the financial impacts on a patient of a particular course of treatment. Looking up the specifics of coverage for a patient requires finding the insurer's provider portal, locating the documents that relate to that patient's insurance plan (i.e., correct market segment) and condition, and then finding the specifics of that provider's payment criteria for the proposed course of treatment. Even when providers can find the correct payment criteria, the differences between these criteria fragment treatment options for patients with similar problems and introduce unnecessary complexity and variability in physician practices.

Worse yet, addressing these byzantine requirements obligates physicians to ensure that their charts contain just the right phrasing needed to support the payment requirements and coding rules of each insurer. Physicians routinely experience interactions where administrators ask them to change something about the way they practice or their documentation in order to increase or expedite payments from insurers. Some in the insurer space call this interaction "physician abrasion", and an Institute of Medicine report from 2010 estimates over 3 hours of physician time per week is consumed by these activities. Similarly, a 2009 Health Affairs article estimated some $23-$31 billion in annual costs from these activities, while other studies suggest that administrative complexity consumes 12% of physicians' patient service revenue. These studies do not address the increased likelihood of a patient receiving a bill for a service they expected to be covered.

Thus, even for "standard claims" that do not require prior authorization, the convoluted nature and poorly organized documentation of payment criteria can make it difficult for a patient or even a provider to know in advance exactly which services will be paid for by that patient's insurer. For patients, the lack of clarity on what will be covered creates significant financial risk, as the patient holds ultimately responsibility to pay the provider for services rendered if the insurer does not. Given that half of Americans hold medical debt, the issue of medical services not covered by insurance is a significant component of the healthcare affordability crisis devastating American consumers

Furthermore, the complexity of these rules impedes any significant shopping by patients, because even when a patient has price information for particular service codes of interest, they cannot really know how the service they receive will be coded, and reimbursed. Despite significant progress on price transparency led by CMS' work on hospital charge masters and transparency in coverage files, if patients cannot readily understand coverage rules, they cannot get real world costs to compare providers. Efforts such as shared decision-making are also much less effective, since the patient cannot begin to understand the relative value of different treatment options without real-world costs.

The maze of rules also impedes shopping for insurance plans. Because of the fragmented and disorganized documentation of payment criteria, patients who need particular services to manage their

health cannot meaningfully look up what the insurer will cover under a plan and what they will need to do to obtain coverage. For example, for a consumerwho is managing diabetes using an insulin pump, coverage of the model of pump they have and compatible insulin are among the most important aspects of plan choice. Yet to find meaningful answers would require the consumer to go through the following steps for each insurer they might be considering:

- understand the existence of payment criteria
- find provider portals for different insurers
- figure out which market segment they are in
- go to the appropriate portion of the manual or provider portal, traverse the differing structure of documentation between each insurer, and finally
- read complex clinical language to understand coverage

In practice consumers, no matter how expert, are unlikely to have the time, expertise and stamina to go through this process for different insurers. With proper organization those documents could be curated for consumer use, but under the current state of affairs it would be impossible for any third party to assemble and track these documents across insurers.

How These Problems Affect Prior Authorization

Prior authorization and other utilization management methodologies, primarily involve making an assessment of whether a particular medical product or service is appropriate for one specific patient as opposed to patients in general. These patient-specific assessments typically extend existing an insurer's all-patient assessment of medical necessity by incorporating information about that specific patient's health status, conditions, and responses to other forms of treatment. Likewise, the payment criteria in prior authorizations frequent build on existing payment criteria, limiting payment to a subset of those patients addressed by "standard claims" payment criteria by requesting extra information from provider beyond what is usually submitted in "standard claims".

As documented through the exhaustive research preceding the proposed rules, it is often difficult and time-consuming to identify the prior auth documentation required by a particular issuer for payment of particular services and even more burdensome to transmit. We note that the poor organization of payment criteria for "standard claims" are a major contributor to the burden of finding coverage information for prior authorizations as well.

We further believe that to be maximally effective resources such as the PAARD API should expose prior authorization and other documentation requirements based on a standardized scheme for documenting both the forms, the information requested, and the underlying criteria. Without standardization of the original payment criteria underlying the requested documentation, providers will struggle to understand the rationale for the prior authorization or evaluate options that may be covered with less wait and burden.

Potential Improvements for Prior Authorization and Insurer Coverage Rules

We suggest that HHS consider undertaking efforts to align:

1- payment criteria themselves, and
2- the ways in which documentation of payment criteria are organized and published.

We would suggest starting by examining payment criteria and actual coding and billing information within claims for the most commonly reimbursed medical services to determine sources of variability. The results of that examination could identify the burden associated with divergent payment rules, as well as providing important information on how these rules affect patient outcomes. Based on the results of the examination, HHS could establish a reasonable set of medical necessity criteria and coding rules for common services, particularly those with low denial rates and lower variability in payment criteria and actual coding practices. A bill before the Massachusetts legislature, HD 3270, provides a useful model for consideration.

HHS should thereafter encourage insurers to adopt these "core" payment criteria across all lines of business. There will likely need to be methods for insurers to opt out or add to these rules; the group should develop both procedures (e.g., documenting and monitoring the reasons for not following the standards) and interoperability standards to avoid re-emergence of fragmented rules. Having such criteria would increase efficiency for insurers and transparency for patients. This type of alignment would be best done in conjunction with a consensus-based entity.

A related work stream would be to propose a general scheme for organizing payment criteria so that they can be consumed in a variety of contexts. For example, the criteria should organized in such a way that a person or app using a reasonably standard definition of an episode could retrieve the coverage criteria for various treatment options within that episode (e.g. different devices, techniques, etc.) and provide a cost comparison. The Good Faith Estimates and Advanced Explanation of Benefits processes required under the No Surprises Act and currently being developed in FHIR workgroups will provide patients with estimates once all the specifics of proposed treatment are established. This is an extraordinary advance for patients. Making available the coverage criteria the underly those cost estimates further empower patients and physicians to fully consider the value (i.e., both costs and benefits) of different courses of action.

We acknowledge the importance of having rules that control the use of medical services both for cost control and in avoiding harms to patients from unnecessary or harmful treatments. For the public writ large, having different insurers perform their own assessment of the value of different interventions serves as an important mechanism for balancing access to novel and potentially valuable medical services with the need for prudent use of these services and cost control.

Nonetheless, for most types of medical encounters and the services that are delivered, it is hard to understand why each insurer must write a completely distinct set of rules. For example, why should insurers maintain their own distinct rules for coding and payment of preventive services recommended by the US Preventive Services task force? In cases like preventive services and many others, distinct sets of payment rules and associated distinct coding do not benefit patients, providers, or the public. Instead, they make it extremely difficult to for patient provider or public form a reasonable expectation of what will be covered in advance of agreeing to use the service.

We note that 29 CFR 2560.503 1(b)(3) requires insurers of group health plans to maintain reasonable claims procedures such that:

> The claims procedures do not contain any provision, and are not administered in a way, that unduly inhibits or hampers the initiation or processing of claims for benefits.

While we are not legal experts, we believe it would not be unreasonable to argue that the current state of documentation of payment criteria “unduly inhibits or hampers the initiation or processing of” numerous claims.